

Retinal-OCT

Abstract

Recent advancements in artificial intelligence have substantially accelerated automated screening, microstructural mapping, and diagnostics in ophthalmology. This literature review synthesizes cutting-edge deep learning (DL) methodologies, convolutional neural networks (CNNs), Vision Transformers (ViTs), and Graph Neural Networks (GNNs) across five core contemporary studies focusing on retinal disease classification, tissue layer segmentation, and spatial risk forecasting. By evaluating diverse diagnostic modalities ranging from 2D color fundus photography to high-resolution 3D Optical Coherence Tomography (OCT) this review details how modern frameworks combine multi-modal, multi-scale, and explainable AI (XAI) techniques.

The synthesized methodologies achieve remarkable performance, with localized multi-disease prediction accuracy reaching up to 98.7%, and graph-fused OCT classifications achieving up to 99.8% accuracy. Key findings emphasize a paradigm shift from uninterpretable, black-box networks to hybrid, geometry-aware, and clinically aligned frameworks that conform to established biological paradigms such as the ISNT rule for glaucoma assessment.

Introduction

The human retina acts as an indispensable window into systemic and ocular health. Conditions such as glaucoma, diabetic retinopathy (DR), age-related macular degeneration (AMD), diabetic macular edema (DME), and cataracts present immense global health challenges, serving as leading causes of irreversible blindness worldwide. Traditional manual diagnostic processes are highly labor-intensive, time-consuming, and subject to severe human error or inter-observer variability, which is further exacerbated by a global shortage of trained ophthalmologists.

To overcome these constraints, deep learning techniques have emerged as powerful tools for automated ocular assessment. While standard CNN backbones excel at isolating high-frequency localized features (e.g., microaneurysms, drusen, and exudates), they frequently lose global contextual information and structural continuity. Conversely, Vision Transformers (ViTs) successfully process long-range pixel dependencies but lack crucial localized architectural inductive biases. Most recently, geometric frameworks specifically Graph Convolutional Networks (GCNs) and Region Adjacency Graphs (RAGs) have bridged this gap by encoding

complex structural topologies, managing spatial configurations, and tracing tissue relationships across varied resolutions.

Synthesis of Core Reviewed Literature

1. Multi-Disease Prediction via Pretrained Baselines

Research focusing on scalable, low-cost screening frameworks leverages transfer learning on large-scale datasets. Using architectures such as pretrained VGG16, VGG19, and ResNet variations, multi-label and multi-task classification layers are trained to differentiate concurrent retinal conditions (such as DR, glaucoma, and cataracts) from standard 2D fundus images. These systems demonstrate that optimizing spatial feature representations can yield localized multi-disease prediction accuracies up to 98.7% and AUC-ROC scores of 0.96.

2. Multi-Resolution Feature Fusion using Graph Networks

To mitigate data heterogeneity across varying scanner manufacturers, hybrid topologies are developed for 3D Optical Coherence Tomography (OCT) image classification. By implementing a Graph-Fused Ensemble combining ResNet50, Inception V3, and ViT-Large with task-specific topology generation, frameworks map structural features across multiple resolutions. This methodology suppresses noisy artifacts while maximizing macro-structural representations, achieving exceptional accuracies of 99.8% on the OCT-2014 dataset, 0.983 on OCTID, and 0.945 on OCTDL.

3. Continuous Layer Segmentation and Glaucoma Detection

Ocular peripapillary OCT scans require strict spatial and topological boundary constraints. Standard pixel-wise U-Nets often output discontinuous tissue labels due to speckle noise or retinal shadowing. To resolve this, modern frameworks integrate a two-stage Convolutional Graph Network (GCN) coupled with a U-shaped architecture to enforce global reasoning across continuous structural boundaries. This geometric approach achieves a Dice Score of 0.850 +/- 0.0012 and a Pixel Accuracy of 0.895 +/- 0.0019, significantly outperforming baseline segmentors like ReLayNet by preserving anatomical continuity.

4. Explainable Graph-Based Spatial Vulnerability Mapping

Addressing the "black-box" dilemma of deep learning in clinical workflows, explainable AI architectures utilize a Disc-Peripheral Dual-Specialist CNN alongside a Superpixel Region

Adjacency Graph (RAG). By computing shortest risk paths via Dijkstra’s routing algorithms, these models transform feature activations into trace-led spatial trajectories. Remarkably, 61.8% of these calculated spatial vulnerability pathways align directly with the biomechanically vulnerable inferior and superior poles of the optic nerve head, mathematically validating the model against the clinical **ISNT rule** (Inferior > Superior > Nasal > Temporal) used by ophthalmologists.

5. Benchmarking Vision Transformers (ViTs) vs. Convolutional Frameworks

Dedicated benchmarking studies systematically contrast standalone Vision Transformers (ViTs), DeepViT, and hybrid ResNet50-ViT architectures for retinal lesion classification. The investigation reveals a critical performance trade-off: while pure ViT models capture long-range global relationships, they struggle with data-scarce medical domains and minute, fine-grained micro-textures (such as focal drusen or early microaneurysms) due to the absence of spatial inductive biases. Conversely, Hybrid CNN-Transformer networks offer optimal diagnostic mapping by maintaining localized convolutional feature filters alongside global self-attention mechanisms.

Comparative Performance Analysis Matrix

Study Focus	Core Architecture / Model	Imaging Modality	Datasets Utilized	Key Performance Metrics
Multi-Disease Screening [1]	Pretrained VGG16 with multi-label & multi-task classification layers	2D Retinal Fundus Images	Kaggle, STAR, DRIVE	Accuracy: 98.7% AUC-ROC: 0.96 Inference Time: ~40 ms
Multi-Resolution Integration [2]	Graph-Fused Ensemble (ResNet50 + Inception V3 + ViT-Large)	Retinal Optical Coherence Tomography (OCT)	OCTDL, OCTID, OCT-2014	OCT-2014 Accuracy: 99.8% OCTID Accuracy: 98.3% OCT-2014 AUC: 1.000

Glaucoma & Tissue Layering [3]	Two-stage Convolutional Graph Network (GCN) + U-Net	Peripapillary OCT scans	Mahatma Eye Hospital Dataset, Duke SD-OCT	Dice Score: 0.850 ± 0.0012\$ Pixel Accuracy: 0.895 ± 0.0019\$ Improvement: +1.9% vs ReLayNet
Explainable Spatial Mapping [4]	Disc-Peripheral Dual-Specialist CNN + Superpixel RAG	2D Retinal Fundus Photography	REFUGE, ORIGA (1,050 total images)	Biological Consistency: 61.8% of risk paths match the inferior/superior vulnerable poles (ISNT rule).
Transformer Benchmarking [5]	Standalone ViT, DeepViT vs. Hybrid ResNet50-ViT Networks	Retinal Optical Coherence Tomography (OCT)	Cross-domain OCT benchmarks	Key Finding: Hybrid models minimize data-scarcity bottlenecks and outperform standalone ViTs on small structural lesions.

Research Gaps

- Scarcity of Longitudinal Prognostics from Static Images:** The vast majority of current state-of-the-art architectures focus purely on static point-in-time classifications. There remains a critical lack of predictive pipelines capable of forecasting a patient's multi-year degenerative trajectory (e.g., progression from early to advanced geographic atrophy) using only a single baseline visit image.
- Naive Multi-Modal and Feature Fusion:** Combining diverse network layers frequently falls back on "naive fusion" (such as basic tensor concatenation), which overloads downstream linear classifiers. Advanced graph alignment layers and cross-attention refinement mechanisms are drastically underutilized.
- Vulnerability to Domain Shifts and Image Artifacts:** Deep models remain highly sensitive to domain shifts caused by multi-vendor imaging hardware, variations in

resolution, optical cataracts, and localized artifacts. Unsupervised domain adaptation and strict topological continuity loss constraints are necessary to prevent localized segmentation fragmentation.

4. **Absence of Tabular Clinical Metadata Integration:** Existing imaging frameworks function in structural isolation. They completely lack geometric pipelines designed to integrate textual clinical markers (patient age, intraocular pressure, HbA1c levels, or genetic risk coefficients) directly into the image-graph topology.

Conclusion

The intersection of deep convolutional backbones, attention-driven transformers, and geometric graph neural networks represents a paradigm shift for automated ophthalmology. Moving away from isolated pixel processing, contemporary frameworks successfully aggregate multi-resolution features, enforce continuous anatomical boundaries, and provide explainable mapping trajectories that directly reflect clinically recognized biological paradigms like the ISNT rule. Future work must bridge remaining translational barriers by incorporating non-imaging clinical metadata, validating cross-vendor generalizability, and engineering longitudinal tracking frameworks capable of forecasting long-term disease progression.

References

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- [4] [*S. R. Kishan, S. S. S. Sathwik, and S. T. S. Balaji, "Explainable Graph-Based Spatial Vulnerability Mapping for Glaucoma Using Fundus Images," 2026*](#)

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Github link: https://github.com/KanizFatema192/Group09_Retinal-OCT_CSE475